

Self similar diffusion

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1 The self-similar framework

1.1 Rescaling into a self-similar frame

Let $u_i^*(\mathbf{x}^*, t^*)$, $i = 1, \dots, d$, satisfy an evolution equation of the form

$$\frac{\partial u_i^*}{\partial t^*} = R_i^*(u^*), \quad (1)$$

on $\mathbb{R}^d$. We rescale amplitude, length, and time by

$$u_i = \frac{u_i^*}{U}, \quad x_j = \frac{x_j^*}{L}, \quad t = \frac{t^*}{s(t)}, \quad (2)$$

where the amplitude scale U and the length scale L depend on the non dimensional time t . The timescale is determined by specifying the clock

$$\tau = \frac{dt^*}{dt}. \quad (3)$$

The clock τ is problem specific and can contain the time dependent length and amplitude scales. It is defined in §4.1 and §5.1 for the two cases considered here.

Throughout, a prime denotes differentiation with respect to t and we use the following abbreviations

$$\sigma_L = \frac{L'}{L}, \quad \sigma_U = \frac{U'}{U}. \quad (4)$$

Differentiating $x_j = x_j^*/L$ with respect to t at fixed x_j^* gives, for each component j ,

$$\frac{dx_j}{dt} = -x_j^* \frac{L'}{L^2} = -x_j \sigma_L. \quad (5)$$

Applying the chain rule to the left-hand side of (1), and using (3) and (5),

$$\begin{aligned} \frac{\partial u_i^*}{\partial t^*} &= \frac{dt}{dt^*} \left(\frac{\partial(Uu_i)}{\partial t} + U \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} \right) \\ &= \frac{U}{\tau} \left(\frac{\partial u_i}{\partial t} + \sigma_U u_i - \sigma_L x_j \frac{\partial u_i}{\partial x_j} \right). \end{aligned} \quad (6)$$

Equating (6) to R_i^* and solving for $\partial u_i / \partial t$ gives the result used throughout this document. Under the rescaling (2), the equation (1) becomes

$$\frac{\partial u_i}{\partial t} = \underbrace{\sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i}_{\text{scaling terms}} + \underbrace{\frac{\tau}{U} R_i^*(Uu, Lx)}_{\text{physical terms}}. \quad (7)$$

Accordingly we split (7) into two flows. The *scaling operator* $\Phi_{\Delta t}^{(S)}$ advances the scaling terms,

$$\Phi_{\Delta t}^{(S)}(u_i) = u_i + \int_t^{t+\Delta t} \left(\sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i \right) dt, \quad (8)$$

and the *physics operator* $\Phi_{\Delta t}^{(P)}$ advances the rest,

$$\Phi_{\Delta t}^{(P)}(u_i) = u_i + \int_t^{t+\Delta t} \frac{\tau}{U} R_i^*(Uu, Lx) dt. \quad (9)$$

Both are written as the field plus the integral of the terms they advance. $\Phi^{(S)}$ is obtained in closed form below and is shared; $\Phi^{(P)}$ is equation-specific and is supplied in §4.2 and §5.6.

1.2 The scaling operator

Evaluating $\Phi_{\Delta t}^{(S)}$ amounts to solving

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i. \quad (10)$$

This is a linear, first-order PDE and can be solved by the method of characteristics. Note that the scaling terms do not couple the components of u_i , so the argument below is componentwise.

Let $x_j(t)$ denote a characteristic curve satisfying

$$\frac{dx_j}{dt} = -x_j \sigma_L. \quad (11)$$

The total derivative of u_i along such a curve is

$$\frac{du_i}{dt} = \frac{\partial u_i}{\partial t} + \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} = \frac{\partial u_i}{\partial t} - \sigma_L x_j \frac{\partial u_i}{\partial x_j}, \quad (12)$$

so along the characteristic (10) reduces to the ordinary differential equation

$$\frac{du_i}{dt} = -\sigma_U u_i. \quad (13)$$

Integrating (11) from t_k to t_{k+1} gives

$$\ln x_j(t_{k+1}) - \ln x_j(t_k) = -[\ln L_{k+1} - \ln L_k] \implies x_j(t_{k+1}) = x_j(t_k) \frac{L_k}{L_{k+1}}, \quad (14)$$

so that the characteristic arriving at the evaluation point $x_j \equiv x_j(t_{k+1})$ originated at

$$x_j(t_k) = x_j \frac{L_{k+1}}{L_k}. \quad (15)$$

Integrating (13) gives

$$u_i(t_{k+1}) = u_i(t_k) U_k / U_{k+1}. \quad (16)$$

Combining this with the characteristic origin (15) and the initial condition $u_i(t_k, \cdot)$ gives the scaling operator in closed form,

$$\Phi_{\Delta t}^{(S)}(u_i)(\mathbf{x}) = \frac{U_k}{U_{k+1}} u_i\left(\mathbf{x} \frac{L_{k+1}}{L_k}\right). \quad (17)$$

Writing the step ratios

$$\tilde{U}_{k+1} = \frac{U_{k+1}}{U_k}, \quad \tilde{L}_{k+1} = \frac{L_{k+1}}{L_k}, \quad (18)$$

this is

$$\Phi_{\Delta t}^{(S)}(u_i)(\mathbf{x}) = \frac{1}{\tilde{U}_{k+1}} u_i\left(\mathbf{x} \tilde{L}_{k+1}\right). \quad (19)$$

The scaling terms in $\mathbf{k}$ -space. For equations formulated spectrally it is useful to have the transform of the scaling terms. Using $\widehat{\cdot}$ for the spatial Fourier transform,

$$\hat{u}_i(\mathbf{k}) = \int u_i(\mathbf{x}) e^{-i\mathbf{k}_j x_j} d\mathbf{x}, \quad (20)$$

and writing the spread term in conservative form,

$$x_j \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j} (x_j u_i) - d u_i, \quad (21)$$

gives

$$\widehat{x_j \frac{\partial u_i}{\partial x_j}} = i k_j \left(i \frac{\partial \hat{u}_i}{\partial k_j} \right) - d \hat{u}_i = - \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + d \hat{u}_i \right), \quad (22)$$

so that the transform of (10) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + d \hat{u}_i \right) - \sigma_U \hat{u}_i. \quad (23)$$

Rather than solving (23) directly in $\mathbf{k}$ -space, it is easier to transform to physical space, apply (19), and transform back; the Fourier transform (20) is linear, so the two are equivalent.

1.3 Spectral interpolation

The closed form (19) requires u_i at the rescaled points $\mathbf{x} \tilde{L}_{k+1}$, which are not grid points. On a uniform periodic grid the represented field is band-limited, so it can be evaluated at arbitrary points exactly, with no interpolation error, by summing its truncated Fourier series there:

$$u_i(\mathbf{x} \tilde{L}_{k+1}) = \sum_{\mathbf{k} \in \mathcal{K}} \hat{u}_i(\mathbf{k}) \exp(i \mathbf{k}_j x_j \tilde{L}_{k+1}), \quad (24)$$

where $\mathcal{K}$ is the set of represented wavenumbers and $\hat{u}_i(\mathbf{k})$ are the normalized transform coefficients of u_i . This is the defining property of the scheme: the scaling step introduces no spatial discretization error of its own.

Queries outside the box. The true problem we are trying to approximate is posed on an infinite box, with u_i going to zero as it approaches the boundaries. For $\tilde{L}_{k+1} > 1$ it is possible that $|x_j \tilde{L}_{k+1}| > H/2$, in which case we would like the rescaling in (24) to return 0. However, this is not the current behavior: instead, it wraps around the periodic box and returns a potentially nonzero value from the other end of the domain. Therefore, to obtain the desired behavior we modify the scaling step to

$$u_i(\mathbf{x} \tilde{L}_{k+1}) = \chi(\mathbf{x} \tilde{L}_{k+1}) \sum_{\mathbf{k} \in \mathcal{K}} \hat{u}_i(\mathbf{k}) \exp(i \mathbf{k}_j x_j \tilde{L}_{k+1}), \quad \chi(\mathbf{y}) = \prod_d \mathbb{I}[|y_d| \leq H/2]. \quad (25)$$

1.4 Constraint functionals

We will set the ratios $\tilde{U}_{k+1}, \tilde{L}_{k+1}$ that the scaling step (19) depends on by requiring the rescaled field to carry prescribed values of two functionals F_1, F_2 of the form

$$F_m \left[\frac{1}{\tilde{U}} u(x \tilde{L}) \right] = \tilde{U}^{-p_m} \tilde{L}^{-q_m} F_m[u], \quad m = 1, 2. \quad (26)$$

The targets F_1, F_2 are fixed once, from the initial condition (§2.2). Given a field u with current values $F_m^* = F_m[u]$, imposing $\tilde{U}^{-p_m} \tilde{L}^{-q_m} F_m^* = F_m$ and taking logarithms gives the following linear system

$$\begin{bmatrix} p_1 & q_1 \\ p_2 & q_2 \end{bmatrix} \begin{bmatrix} \ln \tilde{U} \\ \ln \tilde{L} \end{bmatrix} = \begin{bmatrix} \ln F_1^* - \ln F_1 \\ \ln F_2^* - \ln F_2 \end{bmatrix}. \quad (27)$$

The functionals themselves are equation-specific (§4.3, §5.7); the substep that applies (27) is CONstrained-SCALE (§2.5).

1.5 Power-law ansatz and the step in physical time

The algorithm also needs to *predict* scale ratios over a fraction of a step, at points where no constraint is available to fix them (§2.7). For this we assume that both scales follow a power law in physical time,

$$X = (t^* + \alpha_X)^{\beta_X}, \quad X \in \{U, L\}, \quad (28)$$

so that the ratio over a fraction i of step k is

$$\frac{X_{k+i}}{X_k} = \left(1 + \frac{i \Delta t_k^*}{t_k^* + \alpha_X}\right)^{\beta_X}. \quad (29)$$

Since the time step $\Delta t_k = t_{k+1} - t_k$ is taken in the self-similar time t , the corresponding physical increment must be obtained from the clock,

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau \, dt, \quad (30)$$

which can be approximated with a three-point Adams–Bashforth-style quadrature:

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau \, dt \approx w_0 \tau_k + w_1 \tau_{k-1} + w_2 \tau_{k-2}. \quad (31)$$

The coefficients can be computed as follows.

$$\begin{aligned} w_0 &= \frac{h^3/3 + (2h_1 + h_2)h^2/2 + h_1(h_1 + h_2)h}{h_1(h_1 + h_2)}, \\ w_1 &= -\frac{h^3/3 + (h_1 + h_2)h^2/2}{h_1 h_2}, \\ w_2 &= \frac{h^3/3 + h_1 h^2/2}{h_2(h_1 + h_2)}, \end{aligned} \quad (32)$$

where $h = \Delta t_k$, $h_1 = \Delta t_{k-1}$, $h_2 = \Delta t_{k-2}$.

Eliminating $t_k^* + \alpha_X = X_k^{1/\beta_X}$ from (28) and taking logarithms gives the function that the algorithm uses in both directions,

$$\ln \frac{X_{k+i}}{X_k} = G_X(i, \beta_X; k), \quad G_X(i, \beta; k) = \beta \ln \left(1 + \exp \left(\ln i + \ln \Delta t_k^* - \frac{\ln X_k}{\beta} \right)\right). \quad (33)$$

It is evaluated forward, with $i = \frac{1}{2}$, to predict a half-step ratio (§2.5), and inverted for β , with $i = 1$, to estimate the exponents from the realized history (§2.8).

Logarithmic arithmetic. U_k and L_k may become extremely large or small, so we store $\ln U_k$ and $\ln L_k$ rather than U_k and L_k , and perform all arithmetic on them in the logarithmic number system, where multiplication and division become addition and subtraction, and where sums and differences are evaluated by

$$\log_b(X + Y) = \log_b(X) + \log_b(1 + b^{y-x}), \quad \log_b(X - Y) = \log_b(X) + \log_b(1 - b^{y-x}) \quad (X > Y > 0), \quad (34)$$

with $x = \log_b(X)$, $y = \log_b(Y)$. In particular (31) is evaluated as

$$\ln \Delta t_k^* = \ln(w_0 e^{\ln \tau_k} + w_1 e^{\ln \tau_{k-1}} + w_2 e^{\ln \tau_{k-2}}). \quad (35)$$

2 Algorithm

The integrator below is stated once, for any equation that supplies the three items of §1: a clock τ , a physics operator $\Phi^{(P)}$, and a pair of constraint functionals. Everything else — the splitting, the scale bookkeeping, the exponent estimation — is common.

2.1 Domain, grid, and notation

The spatial domain is periodic with a fixed physical side length H and N grid points per direction, centred at the origin, so that $x_j \in [-H/2, H/2)$ with uniform spacing $\Delta x = H/N$ and grid points

$$x_n = -\frac{H}{2} + n \Delta x, \quad n = 0, 1, \dots, N-1 \quad (36)$$

in each of the d directions. Every integral over the domain below is approximated by the Riemann sum

$$\int f(\mathbf{x}) d\mathbf{x} \approx (\Delta x)^d \sum_{\mathbf{n}} f(\mathbf{x}_{\mathbf{n}}) \quad (37)$$

on this grid.

An index range abbreviates the corresponding tuple of stored values, e.g. $\ln \tau_{k-2:k} = (\ln \tau_{k-2}, \ln \tau_{k-1}, \ln \tau_k)$ and $\Delta t_{k-2:k} = (\Delta t_{k-2}, \Delta t_{k-1}, \Delta t_k)$. The run constants H, N , the time-step constants $\Delta t_{\max}, C, \gamma, \Delta t_{\min}$ of §2.3, the constraint targets F_1, F_2 , and whatever physical parameters the equation carries are visible to every subroutine and are omitted from their signatures. The step Δt_k is updated dynamically. As set out in §1.5, all arithmetic on the scales is done on $\ln U_k, \ln L_k$.

2.2 Initialization

A run is specified by the initial field u_0 , the domain parameters H, N , the time-step constants $\Delta t_{\max}, C, \gamma, \Delta t_{\min}$ of §2.3, a time horizon T_{end} , the splitting scheme $scheme \in \{\text{LIETROTTER}, \text{STRANG}\}$ (§2.6, §2.7), and initial guesses $\beta_U^{(0)}, \beta_L^{(0)}$, which only seed the bracket of the first root-find for β_U, β_L (§2.8). From these:

- The integration loop starts at $t_0 = 0$ and runs while $t_k < T_{\text{end}}$.
- The self-similar frame is initialized at the initial condition itself, $L_0 = U_0 = 1$, i.e. $\ln L_0 = \ln U_0 = 0$.
- The *target* values F_1, F_2 of the constraint functionals (§1.4) are fixed once, from u_0 itself,

$$F_1 = F_1[u_0], \quad F_2 = F_2[u_0]. \quad (38)$$

- The flag β_{ok} , which records whether the current β_U, β_L are usable by STRANG, starts as $\beta_{\text{ok}} = \text{false}$.

2.3 Time step

The step Δt_k is chosen at the start of each step from the field u_k it starts from,

$$\Delta t_k = \min(\Delta t_{\max}, \gamma \Delta t_{k-1}, \Delta t_{\text{CFL}}(u_k; C)), \quad (39)$$

where $\Delta t_{\text{CFL}}(u; C) = \text{MAXSTABLEDT}(u, C)$ (§2.5) is the largest step at which the equation's physics operator runs at the target CFL number C , expressed as a fraction of the time integrator's own stability limit. For Navier–Stokes this is the advective bound (143). The three terms play distinct roles.

- $\Delta t_{\max}$ caps the step when the field is quiet, and is the step of the settled regime.
- Δt_{CFL} shrinks the step when the field concentrates: a momentary high-amplitude event raises $\max |\mathbf{u}|$ and would otherwise carry the physics step past its stability limit.
- $\gamma > 1$ caps *growth only*. The step drops in a single step when an event arrives and recovers geometrically afterwards. Step $k = 0$ has no predecessor and is uncapped. The cap keeps the step ratios $h/h_1, h_1/h_2$ in the quadrature (32) moderate, and it limits the per-step growth of L , which drives the feedback between the predicted half step and the β fit under STRANG (§2.7). We take $C = 0.5, \gamma = 1.1$.

If (39) falls below $\Delta t_{\min}$ the run stops with an error: a runaway $\max |\mathbf{u}|$ should end the run rather than stall it in ever smaller steps. Fixed stepping at $\Delta t_{\max}$ is the special case with no CFL term, $\Delta t_{\text{CFL}} = \infty$. It is what the 1D diffusion equation uses, since its propagator (64) is exact at any step and has no CFL limit to adapt to.

No rejection. Every step is accepted. The CFL is measured on u_k , so the step overshoots its target only if $\max |\mathbf{u}|$ grows within the step; with $C = 0.5$ it has to more than double in a single step to breach the stability limit. Whether that happens is monitored rather than prevented. The *realized* CFL of step k ,

$$\text{CFL}_k = \frac{\Delta t_k}{\Delta t_{\text{CFL}}(u_{k+1}; 1)}, \quad (40)$$

measures the step actually taken against the field it produced. It is recorded every step, and the run warns the first time it exceeds 1. The pre-step value would carry no information whenever the CFL term is active, since it reads back C by construction.

The end of the run. The loop runs while $t_k < T_{\text{end}}$, with $t_{k+1} = t_k + \Delta t_k$, and the last step is *not* shortened to land on T_{end} . A clipped step can be arbitrarily small, which would put an extreme ratio h/h_1 into (32) on the next β update. The run overshoots T_{end} by less than one step, which is immaterial on the logarithmic clock t . For the same reason, output taken at prescribed times t is taken at the first step reaching each time, never by shortening a step.

2.4 Top-level integrator

Algorithm 1 Top-level integrator

Require: $u_0, H, N, \Delta t_{\text{max}}, C, \gamma, \Delta t_{\text{min}}, T_{\text{end}}, \text{scheme}, \beta_U^{(0)}, \beta_L^{(0)}$

- 1: $t_0 \leftarrow 0, k \leftarrow 0$
- 2: $\ln U_0 \leftarrow 0, \ln L_0 \leftarrow 0$
- 3: Compute constraint targets F_1, F_2 from u_0 via (38)
- 4: $\beta_U \leftarrow \beta_U^{(0)}, \beta_L \leftarrow \beta_L^{(0)}, \beta_{\text{ok}} \leftarrow \text{false}$
- 5: Store $(u_0, \ln U_0, \ln L_0, \ln \tau_0, \beta_U, \beta_L, \beta_{\text{ok}})$
- 6: **while** $t_k < T_{\text{end}}$ **do**
- 7: $\Delta t_k \leftarrow (39)$ on u_k ; **error** if $\Delta t_k < \Delta t_{\text{min}}$ ▷ §2.3
- 8: **if** $\text{scheme} = \text{STRANG}$ **and** β_{ok} **then**
- 9: $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{STRANG}(u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L)$ ▷ §2.7
- 10: **else**
- 11: $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{LIETROTTER}(u_k, \ln U_k, \ln L_k, \Delta t_k)$ ▷ §2.6
- 12: **end if**
- 13: $u_{k+1} \leftarrow \text{APPLYBC}(u_{k+1})$ ▷ equation-specific, §2.5
- 14: $\ln \tau_{k+1} \leftarrow \ln \tau(\ln U_{k+1}, \ln L_{k+1})$ ▷ (61), (81)
- 15: $t_{k+1} \leftarrow t_k + \Delta t_k$
- 16: **if** $k + 1 \geq 3$ **then**
- 17: $(\beta_U, \beta_L, \beta_{\text{ok}}) \leftarrow \text{ESTIMATEBETA}(\ln U_{k:k+1}, \ln L_{k:k+1}, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L)$ ▷ §2.8
- 18: **end if**
- 19: Store $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}, \ln \tau_{k+1}, t_{k+1}, \Delta t_k, \text{CFL}_k, \beta_U, \beta_L, \beta_{\text{ok}})$ ▷ (40)
- 20: $k \leftarrow k + 1$
- 21: **end while**

The run parameter *scheme* selects the step function. STRANG needs usable estimates of β_U, β_L , so the loop falls back to LIETROTTER whenever β_{ok} is false. This is always the case for the first three steps $k = 0, 1, 2$: the first β update needs four stored states, so it first runs at the end of step $k = 2$. It is also the case for every step that follows a failed β update (§2.8), until a later update succeeds. The β update itself runs regardless of *scheme*; under *scheme* = LIETROTTER the estimates are diagnostic only.

2.5 Substeps

Both splitting schemes are assembled from the substeps below. None of them applies the boundary treatment, which is handled only by the top-level loop.

Evolve($u, \Delta t_k$). Applies the physics operator $\Phi_{\Delta t_k}^{(P)}$ of (9): it advances the non-scaling terms of (7) over Δt_k , leaving the scales $\ln U, \ln L$ untouched.

MaxStableDt(u, C). Equation-specific: the largest step at which EVOLVE from u runs at the fraction C of its stability limit, the term $\Delta t_{\text{CFL}}(u; C)$ of (39). For Navier–Stokes it is (143). An equation without a CFL limit does not define it and runs at fixed Δt_{max} (§2.3).

Scale($u, \ln \tilde{U}, \ln \tilde{L}$). Applies the scaling operator (19) with prescribed ratios $\tilde{U}, \tilde{L}$, returning $\tilde{U}^{-1}u(\mathbf{x}_n \tilde{L})$ at each grid point, evaluated with the spectral interpolation of §1.3 and set to zero wherever $\mathbf{x}_n \tilde{L}$ leaves the box (25). Here, and wherever the substeps are used, $\tilde{U}$ and $\tilde{L}$ denote the ratio of U , respectively L , after the substep to its value before it. Inside a Lie–Trotter step this is $\tilde{U}_{k+1} = U_{k+1}/U_k$ of (18); inside a Strang step it is a ratio across a half step, $U_{k+1/2}/U_k$ or $U_{k+1}/U_{k+1/2}$ (§2.7).

ConstrainedScale($u, \ln U, \ln L$). Rescales u so that the result carries the fixed targets F_1, F_2 of (38), and advances the log-scale state accordingly:

1. **Functionals of the input**, $F_1^* = F_1[u]$, $F_2^* = F_2[u]$, evaluated on the grid via (37).
2. **Ratios** $\ln \tilde{U}, \ln \tilde{L}$ from the constraint system (27).
3. **Rescale**, $u' = \text{SCALE}(u, \ln \tilde{U}, \ln \tilde{L})$.
4. **Return** u' together with the advanced state $\ln U + \ln \tilde{U}, \ln L + \ln \tilde{L}$.

No time increment enters CONSTRAINEDSCALE: the ratios are fixed by the functionals of the field it is given, not by the elapsed time. The same substep therefore serves as the full scaling step of Lie–Trotter and as the closing scaling half step of Strang.

PredictScale($i, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L$). Predicts the ratios over the fraction i of step k from the power-law form (28) rather than from the constraints. It evaluates $\ln \Delta t_k^*$ from (35), with the weights (32) of the steps $\Delta t_{k-2:k}$ — the last of which is the step about to be taken — and returns $(G_U(i, \beta_U; k), G_L(i, \beta_L; k))$ using (33). The same function G_X is inverted for β in §2.8.

ApplyBC(u). Imposes whatever boundary treatment the equation requires, after the full step has been assembled. Both equations treated here clamp the domain boundary to zero: the 1D diffusion equation its two edge grid points (§4.5), the Navier–Stokes equations the six faces of the box (§5.10), optionally followed by a Leray projection back onto the divergence-free fields (126). For Navier–Stokes the clamp is itself optional: with it disabled APPLYBC is the identity, and the far-field condition is left to the sponge forcing of §5.3, which acts inside EVOLVE.

2.6 LieTrotter: Lie–Trotter splitting

A Lie–Trotter step composes a full physics step with a full scaling step, $u_{k+1} = \Phi_{\Delta t_k}^{(S)}(\Phi_{\Delta t_k}^{(P)}(u_k))$, with the scaling ratios fixed by the constraints. It uses no estimate of β_U, β_L , so it is valid from the first step. It is written for a general input state $(u, \ln U, \ln L)$ and step Δt_k rather than only a stored state $(u_k, \ln U_k, \ln L_k)$, so that STRANG can call it on a half-stepped state.

Algorithm 2 Lie–Trotter step

- 1: **function** LIETROTTER($u, \ln U, \ln L, \Delta t_k$)
 - 2: $u^{(P)} \leftarrow \text{EVOLVE}(u, \Delta t_k)$ ▷ §2.5
 - 3: **return** CONSTRAINEDSCALE($u^{(P)}, \ln U, \ln L$) ▷ §2.5
 - 4: **end function**
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2.7 Strang: Strang splitting

A Strang step is the symmetric composition

$$u_{k+1} = \Phi_{\Delta t_k/2}^{(S)} \left(\Phi_{\Delta t_k}^{(P)} \left(\Phi_{\Delta t_k/2}^{(S)} (u_k) \right) \right) \quad (41)$$

of a scaling half step, a full physics step, and a second scaling half step. The last two of these are exactly a Lie–Trotter step — a full physics step followed by a scaling step whose ratios come from the constraints — so STRANG is a predicted scaling half step followed by a call to LIETROTTER.

Algorithm 3 Strang step

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1: function STRANG( $u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L$ )
2:    $(\ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k)) \leftarrow \text{PREDICTSCALE}(\frac{1}{2}, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L) \triangleright (42)$ 
3:    $u_{k+1/2} \leftarrow \text{SCALE}(u_k, \ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k))$ 
4:    $\ln U_{k+1/2} \leftarrow \ln U_k + \ln(U_{k+1/2}/U_k); \ln L_{k+1/2} \leftarrow \ln L_k + \ln(L_{k+1/2}/L_k)$ 
5:   return LIETROTTER( $u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2}, \Delta t_k$ )  $\triangleright \S 2.6$ 
6: end function
```

No constraint fixes the state at $t_k + \Delta t_k/2$, so the first scaling half step takes its ratios from the current power-law estimates, via PREDICTSCALE with $i = 1/2$:

$$\ln \frac{U_{k+1/2}}{U_k} = G_U(\frac{1}{2}, \beta_U; k), \quad \ln \frac{L_{k+1/2}}{L_k} = G_L(\frac{1}{2}, \beta_L; k). \quad (42)$$

The call to LIETROTTER then advances $u_{k+1/2}$ over the full Δt_k , and its CONSTRAINEDSCALE substep — the second scaling half step — sets $U_{k+1}/U_{k+1/2}$ and $L_{k+1}/L_{k+1/2}$ so that u_{k+1} satisfies the constraints. The half-step state $(u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2})$ is internal to the step and is not stored. The top-level loop calls STRANG only when β_{ok} is true (§2.4), which guarantees that β_U, β_L were fitted to the stored history at the end of the previous step and that $\ln \tau_{k-2}$ exists.

2.8 EstimateBeta: estimating β_U and β_L

After every step with $k+1 \geq 3$, regardless of *scheme*, β_U and β_L are re-estimated from the log-scale history. Four stored states are needed: $\ln X_k$ and $\ln X_{k+1}$ for the ratio, and $\ln \tau_{k-2:k}$ with the steps $\Delta t_{k-2:k}$ for Δt_k^* . The update inverts the function G_X of (33) with $i = 1$, the same function that PREDICTSCALE evaluates forward with $i = 1/2$:

$$G_U(1, \beta_U; k) = \ln U_{k+1} - \ln U_k, \quad (43)$$

and identically with $U \rightarrow L$ for β_L . The steps $\Delta t_{k-2:k}$ end in the step just taken, so the Δt_k^* inverted here is the one PREDICTSCALE used for that step’s half step: under a varying step, prediction and estimation remain exact inverses on the same step.

The sensitivity of β_X to an error in Δt_k^* is weak. Taking logarithms of (33) for a small step, $\ln(\ln X_{k+1} - \ln X_k) \approx \ln \beta_X + \ln \Delta t_k^* - \ln X_k / \beta_X$, so a relative error ϵ in Δt_k^* moves β_X by $\delta \beta_X \approx \beta_X^2 \epsilon / \ln X_k$, damped as the scales grow. It is not zero, however: feeding a stale step into (32) instead of Δt_k makes the estimate oscillate with the step sequence.

Each of (43) and its L -counterpart is a single scalar equation in a single unknown (β_U , resp. β_L), solved by **brentq**; the implementer will need to choose a bracketing interval (e.g. centred on the previous β_U, β_L estimate — on $\beta_U^{(0)}, \beta_L^{(0)}$ for the first update — widened adaptively if it fails to bracket a root).

If both root-finds succeed, ESTIMATEBETA returns the new β_U, β_L with $\beta_{\text{ok}} = \text{true}$. If either fails to bracket a root even after widening, the failed estimate keeps its previous value (which seeds the next bracket), and ESTIMATEBETA returns $\beta_{\text{ok}} = \text{false}$, so the next step falls back to LIETROTTER. Before $k+1 \geq 3$ no update is made: β_U, β_L stay at $\beta_U^{(0)}, \beta_L^{(0)}$ and β_{ok} stays false.

3 Post-processing

A completed run leaves behind the log-scale history $\ln U_k, \ln L_k$ and the clock history $\ln \tau_k$. From these we recover the power-law parameters α_X, β_X of the ansatz (28). This is the offline, global counterpart of ESTIMATEBETA (§2.8), which fits β_X one step at a time; here the whole history is used at once, so the fit also serves as a check on the run.

Write

$$\mathcal{L}_k = \ln L_k, \quad \mathcal{U}_k = \ln U_k, \quad (44)$$

together with the per-step log-ratios

$$\ell_k = \ln L_{k+1} - \ln L_k, \quad \lambda_k = \ln U_{k+1} - \ln U_k, \quad (45)$$

which for a Lie–Trotter step equal $\ln \tilde{L}_{k+1}$ and $\ln \tilde{U}_{k+1}$ of (18), and for a Strang step combine both half-step ratios. The stored scales are their cumulative sums,

$$\mathcal{L}_k = \mathcal{L}_0 + \sum_{i=1}^k \ell_i, \quad \mathcal{U}_k = \mathcal{U}_0 + \sum_{i=1}^k \lambda_i, \quad (46)$$

which grow only linearly in k ; unlike L_k and U_k themselves, $\mathcal{L}_k$ and $\mathcal{U}_k$ never overflow.

3.1 The scaling rates under the ansatz

Differentiating the ansatz (28) with respect to t , and using the clock (3) to convert from t^* , the logarithmic rates (4) are

$$\sigma_X = \frac{1}{X} \frac{dX}{dt^*} \frac{dt^*}{dt} = \frac{\beta_X \tau}{t^* + \alpha_X}, \quad X \in \{U, L\}. \quad (47)$$

Inverting the ansatz to eliminate the unknown offset,

$$t^* + \alpha_X = X^{1/\beta_X}, \quad (48)$$

and substituting into (47) gives the rates in terms of the scales alone,

$$\sigma_X = \beta_X \tau X^{-1/\beta_X}. \quad (49)$$

Taking logarithms,

$$\ln \sigma_{X,k} = \ln \beta_X + \ln \tau_k - \frac{\mathcal{X}_k}{\beta_X}, \quad \mathcal{X}_k = \ln X_k. \quad (50)$$

Finally, the rates themselves are estimated directly from the log-ratios already in hand, without any finite difference of L_k or U_k themselves,

$$\sigma_{L,k} \approx \frac{\ell_k}{\Delta t_k}, \quad \sigma_{U,k} \approx \frac{\lambda_k}{\Delta t_k}, \quad (51)$$

each divided by its own step, so that a run with a varying step (§2.3) is fitted exactly as a fixed-step one. The rates, not the log-ratios ℓ_k, λ_k themselves, are also what the run’s diagnostics plot: a ratio scales with its step, so a CFL-driven dip in Δt_k would read as a dip in the scaling.

3.2 Recovering α_X and β_X

Equation (50) is linear in $\mathcal{X}_k$ once the clock is divided out. Regressing

$$\ln \sigma_{X,k} - \ln \tau_k = \ln \beta_X - \frac{1}{\beta_X} \mathcal{X}_k \quad (52)$$

against $\mathcal{X}_k$ therefore yields β_X twice over: once as $e^{\text{intercept}}$ and once as $-1/\text{slope}$. Agreement between the two is a check that the run is in fact following a power law over the fitted range; disagreement indicates that it is not, or that the range includes the initial transient before the frame has locked on.

With β_X in hand, the offset follows from (48),

$$\alpha_X = X_k^{1/\beta_X} - t_k^*, \quad (53)$$

where the physical time is accumulated from the per-step increments of (31), with the weights (32) of the stored steps (the first two steps, which lack a three-point history, take $\Delta t_i^* \approx \Delta t_i \tau_i$),

$$t_k^* = t_0^* + \sum_{i=0}^{k-1} \Delta t_i^*. \quad (54)$$

In exact arithmetic (53) is independent of k ; in practice it is averaged over the fitted range, and its scatter is a second check on the ansatz.

For the 1D diffusion equation the clock is $\tau = L^2/D$, and (50) specializes to

$$\ln \sigma_{L,k} = \ln \frac{\beta_L}{D} + \left(2 - \frac{1}{\beta_L}\right) \mathcal{L}_k, \quad (55)$$

$$\ln \sigma_{U,k} = \ln \frac{\beta_U}{D} + 2 \mathcal{L}_k - \frac{\mathcal{U}_k}{\beta_U}, \quad (56)$$

so that for $X = L$ the regression (52) collapses onto a single variable, while for $X = U$ it is a two-variable fit in $\mathcal{L}_k$ and $\mathcal{U}_k$.

4 The 1D diffusion equation

The first instantiation of the framework is the one-dimensional diffusion equation, $d = 1$, with a single scalar component; we drop the component index throughout this section. It serves as a testbed: every operator the framework needs is available in closed form, so the only approximation in a run is the operator splitting itself.

4.1 Governing equation and clock

The dimensional equation is

$$\frac{\partial u^*}{\partial t^*} = D \frac{\partial^2 u^*}{\partial (x^*)^2}, \quad (57)$$

so that, in the notation of (1),

$$R^*(u^*) = D \frac{\partial^2 u^*}{\partial (x^*)^2}. \quad (58)$$

Under the rescaling (2) this term becomes

$$D \frac{\partial^2 (uU)}{\partial (x^*)^2} = DU \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{1}{L} \right) \frac{1}{L} = \frac{DU}{L^2} \frac{\partial^2 u}{\partial x^2}, \quad (59)$$

having used $dx/dx^* = 1/L$. The clock is then chosen to make the physical term of the lemma (7) carry unit coefficient,

$$\frac{\tau}{U} \cdot \frac{DU}{L^2} = 1 \quad \implies \quad \tau = \frac{L^2}{D}, \quad (60)$$

which is the diffusive time scale of the current length scale, as expected. In log form, as required by (35),

$$\ln \tau_k = 2 \ln L_k - \ln D. \quad (61)$$

Substituting (60) into the rescaling lemma (7) gives the equation the algorithm integrates,

$$\frac{\partial u}{\partial t} = \sigma_L x \frac{\partial u}{\partial x} - \sigma_U u + \frac{\partial^2 u}{\partial x^2}. \quad (62)$$

4.2 The physics step

Because the diffusion term in (62) has unit coefficient, the physics operator (9),

$$\Phi_{\Delta t}^{(P)}(u_k) = u_k + \int_{t_k}^{t_k + \Delta t} \frac{\partial^2 u}{\partial x^2} dt, \quad (63)$$

has the closed-form spectral solution

$$\hat{u}_j^{(P)} = \hat{u}_j e^{-k_j^2 \Delta t}, \quad k_j = \frac{2\pi j}{H}, \quad j = 0, \dots, \lfloor N/2 \rfloor, \quad (64)$$

where $\hat{u}_j = \text{RFFT}(u)_j$. This is an exact one-shot spectral propagator, not an approximate time-marching scheme, so no sub-stepping or implicit solve is needed, and the accuracy requirement on EVOLVE (§2.5) is met trivially at any order. Inverse-transforming gives the advanced field $u^{(P)}(x_n)$.

4.3 Moment constraints

The constraint functionals of §1.4 are taken to be the zeroth and second moments of the field,

$$F_1[u] = \int u dx \equiv M_0, \quad F_2[u] = \int x^2 u dx \equiv M_2, \quad (65)$$

so that M_0 measures the amount of the conserved quantity and M_2 its spread. Writing M_0^*, M_2^* for the moments of the field being scaled, and substituting $y = x \tilde{L}$ in each integral, the scaled field carries

$$\int \frac{1}{\tilde{U}} u(x \tilde{L}) dx = \frac{1}{\tilde{U} \tilde{L}} M_0^*, \quad (66)$$

$$\int x^2 \frac{1}{\tilde{U}} u(x \tilde{L}) dx = \frac{1}{\tilde{U} \tilde{L}^3} M_2^*, \quad (67)$$

so the homogeneity exponents of (26) are

$$(p_1, q_1) = (1, 1), \quad (p_2, q_2) = (1, 3), \quad (68)$$

with $p_1 q_2 - p_2 q_1 = 2 \neq 0$, so the constraint system (27) is uniquely solvable: the two moments do respond independently to amplitude and to length rescaling. Solving it,

$$\tilde{L} = \sqrt{\frac{M_0 M_2^*}{M_0^* M_2}}, \quad (69)$$

$$\tilde{U} = \frac{M_0^*}{M_0 \tilde{L}} = \left(\frac{M_0^*}{M_0} \right)^{3/2} \sqrt{\frac{M_2}{M_2^*}}. \quad (70)$$

Open. An earlier form of (69) and (70) omitted the factor M_0 , giving $\tilde{L} = \sqrt{M_2^*/(M_0^* M_2)}$ and $\tilde{U} = (M_0^*)^{3/2} M_0^{-1} \sqrt{M_2/M_2^*}$; the present implementation uses that form. The two agree exactly when $M_0 = 1$, and differ by a factor $M_0^{\pm 1/2}$ otherwise. The discrepancy traces to whether the target second moment is taken raw, as in (65) and (38), or normalized by M_0 . Since diffusion conserves M_0 and the initial condition is in practice normalized to unit mass, the difference has not so far been visible in a run. Which convention is intended should be settled, and the implementation brought into line with it.

4.4 Spectral interpolation on the real transform

For $d = 1$ and a real-valued field, the generic interpolation formula (24) is realized on the non-negative frequency components of a real Fast Fourier Transform (RFFT):

$$\begin{aligned} \Phi_{\Delta t}^{(S)}(u_k)(x) = \frac{1}{\tilde{U}_{k+1}} & \left\{ \hat{u}_0^{(k)} + 2 \sum_{j=1}^{N/2-1} (-1)^j \left[\operatorname{Re}(\hat{u}_j^{(k)}) \cos\left(\frac{2\pi j \tilde{L}_{k+1} x}{H}\right) \right. \right. \\ & \left. \left. - \operatorname{Im}(\hat{u}_j^{(k)}) \sin\left(\frac{2\pi j \tilde{L}_{k+1} x}{H}\right) \right] \right. \\ & \left. + (-1)^{N/2} \hat{u}_{N/2}^{(k)} \cos\left(\frac{\pi N \tilde{L}_{k+1} x}{H}\right) \right\}, \end{aligned} \quad (71)$$

where $\hat{u}_j^{(k)} = \frac{1}{N} \text{RFFT}(u_k)_j$ for $j = 0, 1, \dots, N/2$, and the Nyquist term at $j = N/2$ is included only if N is even. The factor of 2 accounts for the Hermitian conjugate modes omitted by the real transform. (71) holds for $|\tilde{L}_{k+1}x| \leq H/2$; outside that the cutoff (25) sets $\Phi_{\Delta t}^{(S)}(u_k)(x) = 0$.

The alternating sign $(-1)^j$ is the phase-origin correction, and it is not optional. The transform phases its coefficients to the sample index n , whose origin $n = 0$ sits at $x = -H/2$ on the centred grid (36) rather than at $x = 0$. Since the scaling operator (19) dilates about the centre of the domain, each mode must be evaluated at $\tilde{L}_{k+1}x + H/2$ — dilate about $x = 0$, then shift to the transform's phase origin — and not at $\tilde{L}_{k+1}(x + H/2)$, which dilates the shift along with the coordinate and agrees with it only at $\tilde{L}_{k+1} = 1$. As $H/2$ is half a period, that shift contributes the constant phase $\exp(ik_j H/2) = (-1)^j$. Dropping it altogether returns $u_k(\tilde{L}_{k+1}x - H/2)$, the wanted profile displaced by half the domain, rather than the $u_k(\tilde{L}_{k+1}x)$ that (19) calls for. The same factor carries to each axis in three dimensions (§5.9).

4.5 Boundary treatment

The substep APPLYBC of §2.5 imposes a homogeneous Dirichlet condition by clamping the two edge grid points of the centred domain (36) after each completed step,

$$u_{k+1}(x_0) \leftarrow 0, \quad u_{k+1}(x_{N-1}) \leftarrow 0. \quad (72)$$

On the centred grid $x \in [-H/2, H/2)$ these are two distinct grid points, not periodic images of one another.

Open. This sits in tension with the rest of the discretization: the spectral propagator (64) presumes a periodic field, whereas (72) imposes a Dirichlet condition. The spectral interpolation (71) no longer does at the edges: with the cutoff (25) it treats everything beyond the box as zero, which is consistent with the clamp, but it is still a periodic interpolant inside the box. The two are consistent only so long as u is negligible near the domain edges, which requires the initial condition to be localized well inside the domain and the run to be stopped before the profile spreads to the boundary. The scaling step works against this — it is what keeps the profile from spreading in the self-similar frame — but the condition is an assumption of the scheme rather than a consequence of it, and is not currently monitored.

5 The Navier–Stokes equations

The second instantiation is the incompressible Navier–Stokes equations, with $d = 3$ and u_i the velocity field. The framework applies unchanged; what this section supplies is the clock, the nondimensional right-hand side, the structure of the physics operator, and the constraint functionals. One item needed to actually run the algorithm of §2 — the time-marching scheme inside $\Phi^{(P)}$ — remains open, and is marked as such below.

5.1 Governing equations and clock

The equations are solved as a large-eddy simulation (LES): u_i^* and p^* denote the filtered (resolved) velocity and pressure, and the effect of the unresolved scales enters through the kinematic subgrid-scale stress τ_{ij}^* .

The dimensional filtered equations are

$$\frac{\partial u_i^*}{\partial t^*} + u_j^* \frac{\partial u_i^*}{\partial x_j^*} = -\frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} + \nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}, \quad (73)$$

together with the continuity equation,

$$\frac{\partial u_i^*}{\partial x_i^*} = 0, \quad (74)$$

so that, in the notation of (1),

$$R_i^*(u^*) = -u_j^* \frac{\partial u_i^*}{\partial x_j^*} - \frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} + \nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}. \quad (75)$$

In addition to the rescaling (2) of u_i and x_j , the pressure is nondimensionalized by the dynamic pressure,

$$p = \frac{p^*}{\rho U^2}, \quad (76)$$

and the subgrid-scale stress, which carries units of velocity squared, by U^2 ,

$$\tau_{ij} = \frac{\tau_{ij}^*}{U^2}. \quad (77)$$

Term by term, the convective, pressure, and subgrid-scale terms scale as U^2/L ,

$$u_j^* \frac{\partial u_i^*}{\partial x_j^*} = \frac{U^2}{L} u_j \frac{\partial u_i}{\partial x_j}, \quad \frac{1}{\rho} \frac{\partial p^*}{\partial x_i^*} = \frac{U^2}{L} \frac{\partial p}{\partial x_i}, \quad \frac{\partial \tau_{ij}^*}{\partial x_j^*} = \frac{U^2}{L} \frac{\partial \tau_{ij}}{\partial x_j}, \quad (78)$$

while the viscous term scales as $\nu U/L^2$,

$$\nu \frac{\partial^2 u_i^*}{\partial x_j^* \partial x_j^*} = \frac{\nu U}{L^2} \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \quad (79)$$

The clock is chosen so that the dominant, advective part of the physical term of the lemma (7) carries unit coefficient,

$$\frac{\tau}{U} \cdot \frac{U^2}{L} = 1 \quad \implies \quad \tau = \frac{L}{U}, \quad (80)$$

which is the eddy turnover time of the current scales. In log form, as required by (35),

$$\ln \tau_k = \ln L_k - \ln U_k. \quad (81)$$

Applying $\tau/U = L/U^2$ to (75) and introducing the instantaneous Reynolds number

$$Re = \frac{UL}{\nu}, \quad (82)$$

the rescaling lemma (7) gives the nondimensional momentum equation

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i - u_j \frac{\partial u_i}{\partial x_j} - \frac{\partial p}{\partial x_i} + \frac{1}{Re} \frac{\partial^2 u_i}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j}, \quad (83)$$

and the continuity equation is unchanged by the rescaling,

$$\frac{\partial u_i}{\partial x_i} = 0. \quad (84)$$

5.2 Subgrid stress model

The subgrid-scale stress τ_{ij}^* of (73) is closed with the standard Smagorinsky model. Its deviatoric part is taken proportional to the resolved strain rate,

$$\tau_{ij}^* - \frac{1}{3}\tau_{kk}^*\delta_{ij} = -2\nu_t^*S_{ij}^*, \quad S_{ij}^* = \frac{1}{2}\left(\frac{\partial u_i^*}{\partial x_j^*} + \frac{\partial u_j^*}{\partial x_i^*}\right), \quad (85)$$

with the isotropic part $\frac{1}{3}\tau_{kk}^*\delta_{ij}$ absorbed into the pressure, and the eddy viscosity

$$\nu_t^* = (C_s\Delta^*)^2|S^*|, \quad |S^*| = \sqrt{2S_{ij}^*S_{ij}^*}, \quad (86)$$

where C_s is the Smagorinsky constant and Δ^* is the filter width. Nondimensionalizing the filter width with the length scale, $\Delta = \Delta^*/L$, which we take equal to the grid spacing, $\Delta = \Delta x$, and noting that $S_{ij}^* = (U/L)S_{ij}$, the eddy viscosity becomes

$$\nu_t^* = UL(C_s\Delta)^2|S|, \quad (87)$$

so that the nondimensional subgrid stress (77) is

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2(C_s\Delta)^2|S|S_{ij}. \quad (88)$$

The stress carries no dependence on U or L : the Smagorinsky model is compatible with the self-similar rescaling. In the limit of infinite Reynolds number, $1/Re = 0$, and the momentum equation (83) becomes

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i - u_j \frac{\partial u_i}{\partial x_j} - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} [2(C_s\Delta)^2|S|S_{ij}], \quad (89)$$

where p now denotes the modified pressure that includes $\frac{1}{3}\tau_{kk}$.

5.3 Sponge forcing

The condition we want at the edge of the domain is that of an unbounded, quiescent far field, $u_i \rightarrow 0$ as $r \rightarrow \infty$, with $r = \sqrt{x_j x_j}$ the distance from the centre of the domain (36). The pseudo-spectral discretization, however, presumes a periodic field, and cannot impose that condition directly. We approximate it instead by a sponge (fringe) layer: a forcing that damps the velocity towards zero in a spherical shell near the edge of the box and vanishes in its interior,

$$f_i = -W(r)u_i. \quad (90)$$

The layer is meant to absorb whatever reaches it without reflecting it. It is not a Brinkman penalization of a solid wall: W ramps up smoothly to a moderate value rather than jumping to a stiff one, and no no-slip condition is imposed at any radius.

The weight function. The weight is the fringe function of Nordström, Nordin & Henningson (1999), applied radially,

$$W(r) = W_{\max} S\left(\frac{r - r_0}{r_1 - r_0}\right), \quad (91)$$

where the smooth step S is

$$S(\xi) = \begin{cases} 0, & \xi \leq 0, \\ \frac{1}{1 + \exp\left(\frac{1}{\xi - 1} + \frac{1}{\xi}\right)}, & 0 < \xi < 1, \\ 1, & \xi \geq 1. \end{cases} \quad (92)$$

The function S is C^∞ on the whole real line. Every derivative of it vanishes at $\xi = 0$ and at $\xi = 1$, it rises monotonically between them, and it is antisymmetric about its midpoint, $S(1 - \xi) = 1 - S(\xi)$, so that $S(1/2) = 1/2$. The weight (91) is therefore exactly zero in the sphere $r \leq r_0$, rises smoothly across the shell $r_0 < r < r_1$, and is exactly $W_{\max}$ for $r \geq r_1$. It has three parameters:

- $W_{\max}$, the damping rate in the fully developed layer, in units of $1/\tau$, i.e. per eddy turnover time (80). Default $W_{\max} = 10$, which damps the velocity by a factor e in a tenth of a turnover. Setting $W_{\max} = 0$ switches the sponge off.
- r_0 , the radius at which the ramp begins. Default $r_0 = 0.35 H$.
- r_1 , the radius at which the ramp ends. Default $r_1 = H/2$.

Smoothness. The sponge must not introduce Gibbs oscillations into the pseudo-spectral evaluation of Wu_i , so W must be smooth not only on the box but on its periodic extension. With $r_1 \leq H/2$ the sphere $r \leq r_1$ lies inside the box, and $W = W_{\max}$ on every face and corner. At $r_1 = H/2$ the sphere touches the centre of each face, where the periodic image just across the face lies at r slightly below $H/2$; since every derivative of S vanishes at $\xi = 1$, the periodic extension of W is still C^∞ there. The default $r_1 = H/2$ is thus the largest admissible value, and it leaves the largest interior. The ramp S climbs more steeply than its nominal width $r_1 - r_0$ suggests, so that width must be resolved by several grid cells: we require $r_1 - r_0 \gtrsim 10 \Delta x$. The defaults give $r_1 - r_0 = 0.15 H$, about $19 \Delta x$ at $N = 128$ and about $10 \Delta x$ at $N = 64$.

The sponge in the self-similar frame. The weight (91) is fixed in the nondimensional coordinates x_j , and W is a rate in units of $1/\tau$. In physical variables (90) is therefore the forcing

$$f_i^* = -\frac{U}{L} W\left(\frac{r^*}{L}\right) u_i^*, \quad (93)$$

since $f_i^* = -(U^2/L) W(r) u_i$ and the lemma (7) multiplies it by $\tau/U = L/U^2$, recovering (90). In physical space the layer thus moves outwards with the length scale L , and its damping rate follows the eddy turnover rate U/L . Like the Smagorinsky stress, the sponge carries no explicit dependence on U or L in the self-similar frame.

The forced equation. Adding (90) to (89), the momentum equation actually solved is

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i - u_j \frac{\partial u_i}{\partial x_j} - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} [2(C_s \Delta)^2 |S| S_{ij}] - W(r) u_i. \quad (94)$$

The forcing is not divergence free: with $\partial_i u_i = 0$, $\partial_i (W u_i) = u_i \partial_i W$, which is nonzero wherever the flow crosses the ramp. It is not required to be. It enters the pressure Poisson equation alongside the convective and subgrid stress terms (§5.5), and the solenoidal projection then removes its gradient part, so the velocity update stays divergence free. The projected forcing is no longer strictly confined to the shell — its gradient part is taken up by the pressure, which is nonlocal — but it remains purely dissipative: P_{il} is an orthogonal projection and u_i is solenoidal, so the rate at which it changes E_0 of (110) is $-2 \int W u_i u_i \, d\mathbf{x} \leq 0$.

Initial condition. The puff must start well inside the undamped sphere $r \leq r_0$, or the sponge will act on it from the first step. With the window (118) of width σ , we take $\sigma \lesssim r_0/4$ as a guideline; it is not enforced.

Scaling drift. The scaling term $\sigma_L x_j \partial_j u_i$ of (94) is an advection with velocity $-\sigma_L \mathbf{x}$. For $\sigma_L > 0$ it points inwards, so the self-similar frame itself carries content away from the sponge, and the layer has only to absorb what the turbulence carries outwards against that drift.

5.4 Fourier representation

Using the Fourier convention (20), the pressure gradient becomes

$$\widehat{\frac{\partial p}{\partial x_i}} = ik_i \hat{p}. \quad (95)$$

The convective term

$$\hat{N}_i = \widehat{u_j \frac{\partial u_i}{\partial x_j}}, \quad (96)$$

will be evaluated pseudo-spectrally, and so will the subgrid stress term. Writing

$$\nu_t = (C_s \Delta)^2 |S| \quad (97)$$

for the nondimensional eddy viscosity of (87), the strain rate is formed spectrally,

$$\hat{S}_{ij} = \frac{i}{2} (k_j \hat{u}_i + k_i \hat{u}_j), \quad (98)$$

the products $|S|$ and $T_{ij} = 2\nu_t S_{ij}$ are formed in physical space, and the divergence is taken spectrally,

$$\hat{G}_i = \frac{\partial \hat{T}_{ij}}{\partial x_j} = ik_j \hat{T}_{ij}. \quad (99)$$

Since $|S|$ is not a polynomial in u , $\hat{G}_i$ carries aliasing that the 2/3 rule does not remove entirely. The sponge forcing (90) is likewise formed as a product in physical space and transformed,

$$\hat{F}_i = \widehat{W u_i}, \quad (100)$$

with W tabulated once on the grid (36). Because W is smooth but not band-limited, $\hat{F}_i$ also carries content beyond the retained band, which the 2/3 mask removes along with that of $\hat{N}_i$ and $\hat{G}_i$. The scaling terms transform as in (23) with $d = 3$, so the transform of (94) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + 3 \hat{u}_i \right) - \sigma_U \hat{u}_i - \hat{N}_i - ik_i \hat{p} + \hat{G}_i - \hat{F}_i, \quad (101)$$

and the transform of the continuity equation (84) is

$$k_i \hat{u}_i = 0. \quad (102)$$

5.5 Pressure and the solenoidal projection

Contracting (101) with k_i and using the continuity constraint gives

$$0 = -\sigma_L \left(k_i k_j \frac{\partial \hat{u}_i}{\partial k_j} + d k_i \hat{u}_i \right) - \sigma_U k_i \hat{u}_i - k_i \hat{N}_i - ik_i^2 \hat{p} + k_i \hat{G}_i - k_i \hat{F}_i. \quad (103)$$

The remaining scaling contribution also vanishes: differentiating (102) with respect to k_j gives

$$\frac{\partial}{\partial k_j} (k_i \hat{u}_i) = \hat{u}_j + k_i \frac{\partial \hat{u}_i}{\partial k_j} = 0 \implies k_i \frac{\partial \hat{u}_i}{\partial k_j} = -\hat{u}_j, \quad (104)$$

so that $k_i k_j \partial \hat{u}_i / \partial k_j = -k_j \hat{u}_j = 0$. Hence the scaling terms are divergence free on their own — the self-similar frame does not generate pressure — and (103) reduces to a Poisson equation for the pressure. Unlike a constant-coefficient viscous term, the subgrid stress is not divergence free, $k_i \hat{G}_i = ik_i k_j \hat{T}_{ij} \neq 0$ in general, and neither is the sponge forcing, $k_i \hat{F}_i = -i \widehat{u_i \partial_i W}$ (§5.3), so both enter the source alongside the convection,

$$ik^2 \hat{p} = -k_i \left(\hat{N}_i - \hat{G}_i + \hat{F}_i \right) \implies \hat{p} = i \frac{k_l \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right)}{k^2}, \quad k \neq 0, \quad (105)$$

with $\hat{p}$ at $k = 0$ fixed arbitrarily, since only its gradient enters the momentum equation. Substituting (105) into (101), the pressure term becomes

$$-ik_i \hat{p} = \frac{k_i k_l}{k^2} \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right), \quad (106)$$

which combines with $-\hat{N}_i + \hat{G}_i - \hat{F}_i$ to give the solenoidal projection of the convective, subgrid stress, and sponge terms together. Defining the projection tensor

$$P_{il}(\mathbf{k}) = \delta_{il} - \frac{k_i k_l}{k^2}, \quad (107)$$

the evolution equation splits exactly along the lines of the lemma (7),

$$\frac{\partial \hat{u}_i}{\partial t} = \underbrace{-\sigma_L \left(k_j \frac{\partial \hat{u}_i}{\partial k_j} + 3 \hat{u}_i \right)}_{\text{scaling terms}} - \underbrace{\sigma_U \hat{u}_i - P_{il}(\mathbf{k}) \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right)}_{\text{physical terms}}. \quad (108)$$

5.6 The physics step

The physics operator (9) is the flow of the bracketed physical terms of (108),

$$\Phi_{\Delta t}^{(P)}(\hat{u}_i(t)) = \hat{u}_i(t) - \int_t^{t+\Delta t} P_{il}(\mathbf{k}) \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right) dt. \quad (109)$$

The sponge is thus part of the physics operator, not a separate substep: it is switched off by setting $W_{\max} = 0$, independently of APPLYBC (§5.10).

Unlike the 1D case, (109) has no closed form: the convective and subgrid stress terms are both nonlinear, so $\Phi^{(P)}$ must be realized by an actual time-marching scheme. Two things constrain the choice. It must meet the accuracy requirement of §2.5 — second order for the Strang composition to retain its order — and it must handle the dealiasing of the pseudo-spectral products (96) and (99).

We take $\Phi^{(P)}$ to be a single step of the classical fourth-order Runge–Kutta method (RK4) applied to the full physical right-hand side, the subgrid stress term $\hat{G}_i$ and the sponge forcing $\hat{F}_i$ marched together with the convection. It is set out in Appendix A, together with the stability limits it places on Δt . The advective limit depends on the state, and is enforced by the CFL time step of §2.3, through MAXSTABLEDT (143). The sponge limit (142) is fixed by $W_{\max}$ and is assumed rather than enforced.

5.7 Constraint functionals

The moments (65) used for the 1D equation do not carry over: a velocity field in a periodic box has $\int u_i d\mathbf{x} \approx 0$, so F_1 as written is not usable as a scale. We therefore take the corresponding moments of the energy density $u_i u_i$, which is non-negative and so cannot cancel,

$$F_1[u] = \int u_i u_i d\mathbf{x} \equiv E_0, \quad F_2[u] = \int r^2 u_i u_i d\mathbf{x} \equiv E_2, \quad (110)$$

where $r^2 = x_j x_j$ is the squared distance from the centre of the domain (36). Thus E_0 is twice the total kinetic energy and E_2/E_0 the energy-weighted mean square radius, so E_0 measures the amplitude of the field and E_2 its spread, exactly as M_0 and M_2 do in the 1D case.

Writing E_0^*, E_2^* for the values carried by the field being scaled and substituting $\mathbf{y} = \mathbf{x} \tilde{L}$ in each integral, with $d = 3$,

$$\int \frac{1}{\tilde{U}^2} u_i u_i(\mathbf{x} \tilde{L}) d\mathbf{x} = \frac{1}{\tilde{U}^2 \tilde{L}^3} E_0^*, \quad (111)$$

$$\int r^2 \frac{1}{\tilde{U}^2} u_i u_i(\mathbf{x} \tilde{L}) d\mathbf{x} = \frac{1}{\tilde{U}^2 \tilde{L}^5} E_2^*, \quad (112)$$

so the homogeneity exponents of (26) are

$$(p_1, q_1) = (2, 3), \quad (p_2, q_2) = (2, 5), \quad (113)$$

with $p_1 q_2 - p_2 q_1 = 4 \neq 0$, so the constraint system (27) is uniquely solvable. Solving it,

$$\tilde{L} = \sqrt{\frac{E_0 E_2^*}{E_0^* E_2}}, \quad (114)$$

$$\tilde{U} = \left(\frac{E_0^*}{E_0}\right)^{1/2} \tilde{L}^{-3/2} = \left(\frac{E_0^*}{E_0}\right)^{5/4} \left(\frac{E_2}{E_2^*}\right)^{3/4}. \quad (115)$$

The length ratio (114) is identical in form to its 1D counterpart (69): in both cases $\tilde{L}$ is the square root of a ratio of mean square widths.

Open. Two properties of this pair are assumptions rather than consequences. First, the homogeneity (26) is exact only for integrals over $\mathbb{R}^d$; on the finite box the substitution $\mathbf{y} = \mathbf{x} \tilde{L}$ rescales the domain of integration as well, so (111) and (112) hold only to the extent that the energy is concentrated well inside the box. This is the same assumption the 1D moments make, and like it is not currently monitored. Second, r is measured from a fixed origin and is not periodic, so E_2 is meaningful only for a field that stays centred and localized: for a statistically homogeneous field filling the box, $u_i u_i$ is uniform on average and E_2/E_0 reduces to the mean square radius of the box itself, carrying no information about the field. Which of these two regimes the intended runs occupy should be settled before the pair is used.

5.8 Initial condition

Because the curl of a vector field is divergence-free we will define u_i as the curl of some field A_i

$$u_i = \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j}, \quad \text{equivalently} \quad \hat{u}_i = i \varepsilon_{ijk} k_j \hat{A}_k. \quad (116)$$

u_i is built on the grid (36) in four steps, followed by a rescaling.

1. *Noise.* Each component $A_k(\mathbf{x}_n)$ is an independent standard normal, drawn from a seeded Mersenne Twister.
2. *Spectral filter.* The noise is given a spectrum,

$$\hat{A}_k \leftarrow \hat{A}_k e^{-k^2/k_p^2}. \quad (117)$$

3. *Window.* The potential is localized in physical space,

$$A_k \leftarrow A_k e^{-r^2/2\sigma^2}, \quad (118)$$

with r the same distance from the centre of the domain that E_2 uses in (110).

4. *Rescaling.* Finally $u_i \leftarrow u_i \sqrt{E_0/F_1[u]}$, with F_1 evaluated as the Riemann sum (37), so that the field delivered to the integrator carries a prescribed E_0 . The constraint targets (38) are then $F_1 = E_0$ by construction and $F_2 = E_2[u_0]$, the latter set by σ rather than chosen independently.

5.9 Spectral interpolation in three dimensions

The scaling operator (19) needs the same band-limited evaluation (24) as in one dimension, but at $d = 3$ a direct evaluation of it is no longer affordable, and its realization on the real transform differs enough from (71) to be worth writing out.

The sum on the real transform. Write $\hat{u}_i^{(k)}(\mathbf{j}) = N^{-3} \text{RFFT}(u_k)_\mathbf{j}$ for the normalized coefficients, with the first axis carrying only the non-negative frequencies $j_1 = 0, 1, \dots, N/2$ and the remaining two carrying the full signed set $j_d = 0, 1, \dots, N/2, -(N/2 - 1), \dots, -1$. Two corrections turn (24) into a sum over that layout. First, the omitted Hermitian conjugates are restored by the multiplicity

$$m_{j_1} = \begin{cases} 1, & j_1 = 0, \text{ or } j_1 = N/2 \text{ with } N \text{ even,} \\ 2, & \text{otherwise,} \end{cases} \quad (119)$$

exactly as the factor of 2 does in (71). Second, the transform phases its coefficients to the sample index n , which on the centred grid (36) places its origin at $x = -H/2$ rather than at $x = 0$. Since the scaling operator dilates about the centre of the domain, each mode must be evaluated at $\tilde{L}x + H/2$, not at $\tilde{L}(x + H/2)$ — the two agree only at $\tilde{L} = 1$. As $H/2$ is half a period, the correction is the constant phase $\exp(ik_j H/2) = (-1)^j$ per axis. Together,

$$\Phi_{\Delta t}^{(S)}(u_k)_i(\mathbf{x}) = \frac{1}{\tilde{U}_{k+1}} \text{Re} \sum_{j_1=0}^{N/2} \sum_{j_2} \sum_{j_3} m_{j_1} \hat{u}_i^{(k)}(\mathbf{j}) \prod_{d=1}^3 (-1)^{j_d} \exp\left(\frac{2\pi i j_d \tilde{L}_{k+1} x_d}{H}\right). \quad (120)$$

Separable evaluation. Evaluated as written, (120) costs $O(N^{2d})$: every retained mode is summed at every grid point, which at $N = 128$ is 4×10^{12} complex multiply-adds per scaling substep. The kernel, however, is a product of one factor per axis, so the triple sum separates and can be applied one axis at a time. Define the two interpolation matrices

$$W_{nj_1}^{(1)} = m_{j_1} (-1)^{j_1} \exp\left(\frac{2\pi i j_1 \tilde{L}_{k+1} x_n}{H}\right), \quad W_{nj}^{(f)} = (-1)^j \exp\left(\frac{2\pi i j \tilde{L}_{k+1} x_n}{H}\right), \quad (121)$$

the first of size $N \times (N/2 + 1)$ for the half-spectrum axis and the second of size $N \times N$ for the two full axes, both rebuilt whenever $\tilde{L}_{k+1}$ changes. Then (120) is the sequence of contractions

$$\hat{u}_i \xrightarrow{W^{(f)} \text{ on axis 3}} \xrightarrow{W^{(f)} \text{ on axis 2}} \xrightarrow{W^{(1)} \text{ on axis 1}} \text{Re}(\cdot), \quad (122)$$

each of which is a matrix-matrix product once the contracted axis is brought to the front. The cost falls to $O(dN^{d+1})$, i.e. 8×10^8 multiply-adds at $N = 128$, a factor of 5×10^3 below the direct sum, and the work is expressed as three dense GEMM calls. The factorization is exact — it is a regrouping of the same finite sum, not an approximation — so the guarantee of §1.3 that the scaling step contributes no spatial discretization error is retained. For $d = 1$ the sequence (122) collapses to the single contraction with $W^{(1)}$, which is (71).

The Nyquist bin. For even N the bins $j_d = \pm N/2$ of a real field's transform coincide, and the two axis layouts resolve that differently. On the half-spectrum axis the mode is stored once and is given multiplicity $m_{N/2} = 1$ by (119). On a full axis it is stored in the single bin $j_d = N/2$ but represents the conjugate pair, so it must be split evenly between $+N/2$ and $-N/2$; the two halves combine to

$$W_{n, N/2}^{(f)} = (-1)^{N/2} \cos\left(\frac{\pi N \tilde{L}_{k+1} x_n}{H}\right), \quad (123)$$

the same cosine that closes (71). With (123) in place the contraction (122) reproduces the direct sum (120) to roundoff for either parity of N .

The out-of-box cutoff. The box indicator χ of (25) is a product of one factor per axis, like the kernel, so it separates the same way. It is imposed by zeroing row n of each interpolation matrix (121) whenever the query of that axis leaves the box,

$$W_{nj_1}^{(1)} \leftarrow 0, \quad W_{nj}^{(f)} \leftarrow 0 \quad \text{whenever } |x_n \tilde{L}_{k+1}| > H/2. \quad (124)$$

After the three contractions (122) the point $\mathbf{x}_n$ then carries the product of the three row factors, which is $\chi(\mathbf{x}_n \tilde{L}_{k+1})$ exactly. The cutoff is thus applied at no cost beyond building the matrices, and the direct sum it is tested against applies χ pointwise. For $\tilde{L}_{k+1}$ within a few percent of unity the zeroed rows are the outermost one or two grid points per side. These lie deep in the sponge layer (§5.3), where the field has already been damped, and APPLYBC, when enabled, clamps most of them anyway (§5.10). Zeroing a region in physical space is not a solenoidal operation, so, like the clamp, it reintroduces a divergence proportional to the amplitude it removes.

Interaction with dealiasing. The dilation maps each represented wavenumber to $\tilde{L}_{k+1}\mathbf{k}$, so a field that fills the spectrum up to the Nyquist wavenumber would wrap under any $\tilde{L}_{k+1} > 1$. The 2/3 rule already required by the pseudo-spectral product (96) supplies the margin: the top retained mode sits at $N/3$ and moves to $\tilde{L}_{k+1}N/3$, which stays below $N/2$ for every $\tilde{L}_{k+1} < 3/2$. Runs in the self-similar frame hold $\tilde{L}_{k+1}$ within a few percent of unity, so the margin is not approached, but it is the condition under which the interpolation stays free of aliasing. Finally, (122) is applied to each velocity component independently and mixes neither components nor the wavenumbers within an axis; at the exceptional value $\tilde{L}_{k+1} = 1$ it reduces to the plain inverse transform, which is taken instead at $O(N^d \log N)$.

5.10 Boundary treatment

Two boundary treatments are available, and each can be switched on or off independently of the other. The first is the sponge forcing of §5.3, which is part of the physics step and is switched off by $W_{\max} = 0$. The second is the clamp described below, applied by APPLYBC after each step and switched off by a run flag, in which case APPLYBC is the identity. The clamp was the original treatment; the sponge is intended to replace it. The intended configuration is therefore sponge on, clamp off, and the clamp is retained so that the two can be compared: if the sponge alone keeps u_i negligible at the faces, the clamp, and the divergence it introduces, are no longer needed.

The clamp. When enabled, APPLYBC imposes a homogeneous Dirichlet condition on the velocity by clamping the outer layer of grid points of the centred domain (36) after each completed step,

$$u_i(\mathbf{x}_n) \leftarrow 0 \quad \text{whenever } n_d \in \{0, N-1\} \text{ for some direction } d, \quad (125)$$

for every component $i = 1, 2, 3$. As in one dimension, on the centred grid $x_d \in [-H/2, H/2)$ the layers $n_d = 0$ and $n_d = N-1$ are distinct grid points rather than periodic images of one another, so both are clamped; the six layers together are the boundary of the box.

Open. The clamp carries the tension of §4.5 over to the Navier–Stokes case, and sharpens it. As in one dimension, the discretization presumes periodicity — the pseudo-spectral evaluation of (108) does, and the spectral interpolation (120) does inside the box, though its cutoff (124) returns zero outside it — while (125) imposes a Dirichlet condition, and the two are consistent only so long as u_i is negligible near the faces. Beyond that, the clamp is not a solenoidal perturbation: (102) is enforced in Fourier space, and zeroing a layer of grid points in physical space reintroduces a divergence, so the field delivered to the next step no longer satisfies the continuity equation exactly. The size of that violation is set by the amplitude the clamp removes, and therefore by the same localization assumption the constraint pair of §5.7 already makes: for a puff held several widths inside the box it is at roundoff, and it grows with the fraction of E_0 the boundary layer carries. With the sponge on, that fraction is itself suppressed by the damping in the layer, so the clamp has little left to remove.

Optional Leray projection. The violation can instead be removed outright. With the Leray option enabled, APPLYBC follows the clamp (125) by projecting the clamped field back onto the solenoidal subspace with the projection tensor (107),

$$\hat{u}_i(\mathbf{k}) \leftarrow P_{il}(\mathbf{k}) \hat{u}_l(\mathbf{k}) = \hat{u}_i - \frac{k_i k_l \hat{u}_l}{k^2}, \quad \mathbf{k} \neq \mathbf{0}, \quad (126)$$

and transforming back to physical space. The mean mode $\mathbf{k} = \mathbf{0}$ is left unchanged, and the Nyquist bins — which have no representable derivative, and on which P_{il} does not respect the conjugate symmetry of the

real transform — are set to zero. The 2/3 dealiasing mask is not applied here: the next physics step dealias its input anyway (§5.6), and applying the mask here would remove far more than the clamp changed. After (126) the discrete divergence $k_i \hat{u}_i$ is at roundoff, and because P_{il} is an orthogonal projection it can only lower E_0 .

What this gives up is the clamp itself. P_{il} is nonlocal in physical space: the gradient part it removes is the gradient of a potential ϕ solving $\nabla^2 \phi = \nabla \cdot \mathbf{u}$, and that potential is spread over the whole box. So the boundary layers of (125) come back out of (126) with small nonzero values, and a slowly decaying potential-flow tail is added to the field — the same effect §5.8 notes for projecting a windowed initial condition. The two treatments are therefore complementary: the clamp alone keeps the faces exactly at zero and allows a divergence proportional to the amplitude it removes, while the clamp followed by (126) keeps the field exactly solenoidal and lets the faces move away from zero by the same order. Both are small only when u_i is negligible near the faces, so the localization monitored by `leakage` is still needed either way. The option is off by default.

5.11 Dissipation rate post-processing

With $1/Re = 0$ all dissipation is subgrid: the rate at which kinetic energy is transferred from resolved to subgrid scales is

$$\epsilon = -\tau_{ij}^* S_{ij}^* = 2\nu_t^* S_{ij}^* S_{ij}^* = \nu_t^* |S^*|^2, \quad (127)$$

where only the deviatoric part (85) of τ_{ij}^* contributes, since $S_{kk}^* = 0$. Writing this in nondimensional variables with (87) and $S_{ij}^* = (U/L) S_{ij}$,

$$\epsilon = [UL(C_s \Delta)^2 |S|] \frac{U^2}{L^2} |S|^2 = \frac{U^3}{L} (C_s \Delta)^2 |S|^3, \quad (128)$$

evaluated pointwise from the spectrally computed S_{ij} of (98). We are interested in the dissipation rate at the origin, $\mathbf{x}^* = \mathbf{x} = 0$, which is a fixed point of the rescaling (2) and a grid point of (36) ($n = N/2$ for even N), so no interpolation is required. Substituting the power-law ansatz (28) for U and L ,

$$\epsilon(0, t^*) = \frac{(t^* + \alpha_U)^{3\beta_U}}{(t^* + \alpha_L)^{\beta_L}} (C_s \Delta)^2 |S(0, t)|^3. \quad (129)$$

Scaling of the ensemble average. Let $\bar{\cdot}$ denote the average over an ensemble of initial conditions. We make two assumptions:

1. The parameters $\alpha_U, \beta_U, \alpha_L, \beta_L$ are the same for every member of the ensemble, i.e. $U(t^*)$ and $L(t^*)$ are invariant to the initial condition. Then the clock $\tau = L/U$ is too, and so is the map between physical and self-similar time, $t(t^*) = \int_0^{t^*} ds/\tau(s)$; averaging at fixed t^* is therefore the same as averaging at fixed t , and the prefactor of (129) comes out of the average.
2. In the self-similar frame the flow is statistically stationary, so that $\overline{|S(0, t)|^3}$ is independent of t . This is the content of the flow being self-similar; without it $|S|^3$ would contribute its own time dependence to the exponent below.

Under these assumptions

$$\bar{\epsilon}(0, t^*) = \frac{(t^* + \alpha_U)^{3\beta_U}}{(t^* + \alpha_L)^{\beta_L}} (C_s \Delta)^2 \overline{|S(0)|^3}, \quad (130)$$

where C_s and Δ are constants of the run. For $t^* \gg \max(|\alpha_U|, |\alpha_L|)$ the ensemble-averaged dissipation rate therefore follows the power law

$$\bar{\epsilon}(0, t^*) \sim (t^*)^{\gamma_\epsilon}, \quad \gamma_\epsilon = 3\beta_U - \beta_L. \quad (131)$$

Numerical experiment. To test (131) we run an ensemble of M simulations, $m = 1, \dots, M$, that differ only in the seed of the noise in step 1 of §5.8; the parameters k_p, σ, E_0 , the grid, and the constants C_s and Δ are held fixed, so that every member carries the same constraint targets. For each run the following are recorded at every step k :

- the physical time $t_k^{*(m)}$, accumulated as in (54) with a common origin $t_0^* = 0$;
- the dissipation rate at the origin, stored as its logarithm to avoid overflow (§1.5),

$$\ln \epsilon_k^{(m)} = 3 \ln U_k - \ln L_k + 2 \ln(C_s \Delta) + 3 \ln |S(0, t_k)|; \quad (132)$$

- the fitted parameters $\alpha_X^{(m)}, \beta_X^{(m)}$ of §3.2.

Since each run takes its own steps, the samples $t_k^{*(m)}$ do not coincide across the ensemble. Each history is therefore interpolated, linearly in $\ln \epsilon$ against $\ln t^*$, onto a common logarithmically spaced grid t_j^* covering the range reached by every run, and the ensemble mean and its standard error are formed there,

$$\bar{\epsilon}_j = \frac{1}{M} \sum_{m=1}^M \epsilon^{(m)}(t_j^*), \quad s_j = \left[\frac{1}{M(M-1)} \sum_{m=1}^M (\epsilon^{(m)}(t_j^*) - \bar{\epsilon}_j)^2 \right]^{1/2}. \quad (133)$$

The average is of ϵ itself, not of $\ln \epsilon$, to match (130); the sums are evaluated in the logarithmic number system (34).

A power law with an offset,

$$\ln \bar{\epsilon}_j = \ln A + \hat{\gamma}_\epsilon \ln(t_j^* + \alpha_\epsilon), \quad (134)$$

is then fitted to the mean by weighted nonlinear least squares in $(\ln A, \hat{\gamma}_\epsilon, \alpha_\epsilon)$, with weights $(\bar{\epsilon}_j/s_j)^2$ (the inverse variance of $\ln \bar{\epsilon}_j$), over a range $t_j^* \geq t_{\min}^*$ that excludes the initial transient. A confidence interval on $\hat{\gamma}_\epsilon$ is obtained by bootstrap resampling of the M runs. The fitted exponent is compared with the prediction (131), evaluated with the ensemble means $\bar{\beta}_U, \bar{\beta}_L$ of the per-run fits,

$$\gamma_\epsilon = 3\bar{\beta}_U - \bar{\beta}_L, \quad (135)$$

whose uncertainty follows from the spread of $\beta_U^{(m)}, \beta_L^{(m)}$. The spread of $\alpha_X^{(m)}, \beta_X^{(m)}$ across the ensemble is itself a direct check on assumption 1.

Beyond the asymptotic exponent, (130) predicts the whole curve up to the single constant $(C_s \Delta)^2 |S(0)|^3$. We therefore also plot, against t^* , the compensated dissipation rate

$$\bar{\epsilon}_j \frac{(t_j^* + \bar{\alpha}_L)^{\bar{\beta}_L}}{(t_j^* + \bar{\alpha}_U)^{3\bar{\beta}_U}}, \quad (136)$$

which should be flat and equal to $(C_s \Delta)^2 |S(0)|^3$ once the transient has passed. A residual trend in (136) indicates that one of the two assumptions fails — most likely that $|S(0, t)|^3$ has not become stationary in the self-similar frame. The raw curve $\bar{\epsilon}_j$, the fit (134), and the prediction (130) are shown together on logarithmic axes, with the reference slope γ_ϵ of (135).

A The RK4 physics step

This appendix sets out the scheme used for the Navier–Stokes physics operator $\Phi^{(P)}$ of §5.6: one step of the classical fourth-order Runge–Kutta method. Throughout we abbreviate the step as $h \equiv \Delta t$, reserving Δ for the filter width of the subgrid stress model of §5.2.

A.1 The scheme

Written out mode by mode, the physics flow (109) is the system of ordinary differential equations

$$\frac{d\hat{u}_i}{dt} = R_i(\hat{u}), \quad R_i(\hat{u}) = -P_{il}(\mathbf{k}) \left(\hat{N}_l - \hat{G}_l + \hat{F}_l \right), \quad (137)$$

the bracketed physical terms of (108). RK4 advances it through four evaluations of R ,

$$\begin{aligned} \hat{k}_i^{(1)} &= R_i(\hat{u}^{(n)}), & \hat{k}_i^{(2)} &= R_i(\hat{u}^{(n)} + \frac{h}{2}\hat{k}^{(1)}), \\ \hat{k}_i^{(3)} &= R_i(\hat{u}^{(n)} + \frac{h}{2}\hat{k}^{(2)}), & \hat{k}_i^{(4)} &= R_i(\hat{u}^{(n)} + h\hat{k}^{(3)}), \end{aligned} \quad (138)$$

combined with Simpson weights,

$$\hat{u}_i^{(n+1)} = \hat{u}_i^{(n)} + \frac{h}{6} \left(\hat{k}_i^{(1)} + 2\hat{k}_i^{(2)} + 2\hat{k}_i^{(3)} + \hat{k}_i^{(4)} \right). \quad (139)$$

The convective part of each evaluation is computed pseudo-spectrally in rotational form, $u_j \partial_j u_i = -(\mathbf{u} \times \boldsymbol{\omega})_i + \partial_i(u_k u_k/2)$, whose gradient part P_{il} annihilates; this costs six inverse and three forward transforms. The subgrid stress term (99) costs six further inverse transforms, one per independent component of S_{ij} , and six forward transforms of T_{ij} . The sponge term (100) reuses the physical-space velocity already formed for the convective term and multiplies it by the tabulated W , so it costs only the three forward transforms of $W u_i$. An evaluation thus costs 24 transforms and a step 96 in all. The scheme has no coefficients beyond the four weights, so nothing is precomputed or cached.

Order. The scheme is fourth order in h , so it satisfies the second-order requirement §2.5 places on EVOLVE for the Strang composition with a margin.

A.2 Stability

Applied to the linear model $d\hat{u}/dt = \lambda \hat{u}$, (138)–(139) multiply $\hat{u}$ by the amplification factor

$$A(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}, \quad z = \lambda h, \quad (140)$$

the degree-four Taylor polynomial of e^z , and the step is stable while $|A(z)| \leq 1$.

The advective limit. The convective term is, locally, a transport at speed $|\mathbf{u}|$, whose eigenvalues on the retained band lie on the imaginary axis with $|\lambda| \leq \max |\mathbf{u}| k_{\max}$, where $k_{\max} = 2\pi(N/3)/H$ is the top mode retained by the 2/3 rule. Along the imaginary axis $|A(z)| \leq 1$ holds for $|z| \leq 2\sqrt{2}$, so

$$h \max |\mathbf{u}| k_{\max} \leq 2\sqrt{2}. \quad (141)$$

The self-similar frame pins $\max |\mathbf{u}| = O(1)$ on average, so the limit does not drift over a run as it would in a decaying simulation in physical variables. It does not pin it pointwise in time, however: the field can momentarily concentrate into a high-amplitude structure whose $\max |\mathbf{u}|$ carries a step that is stable in the settled regime past (141).

The sponge limit. The sponge forcing (90) is, pointwise, a linear decay at rate $W(r) \in [0, W_{\max}]$, whose eigenvalues lie on the negative real axis with $|\lambda| \leq W_{\max}$. Along the negative real axis $|A(z)| \leq 1$ holds for $-2.785 \lesssim z \leq 0$, so

$$h W_{\max} \leq 2.785. \quad (142)$$

Unlike (141), this limit does not depend on the state: it is fixed by the run parameter $W_{\max}$. It is not enforced by MAXSTABLEDT, but assumed. At the default $W_{\max} = 10$ and a CFL-limited step of $h \approx 0.05$, $h W_{\max} \approx 0.5$, well inside both (142) and the range $h W_{\max} \lesssim 1$ in which the step also follows the exponential decay e^{-Wh} accurately. A run with a larger $W_{\max}$ or a longer step must check (142) by hand.

The CFL time step. The limit is therefore enforced, by the step rule (39) of §2.3. Its CFL term, MAXSTABLEDT, is the step that puts (141) at the fraction C of its limit,

$$\Delta t_{\text{CFL}}(u; C) = C \frac{2\sqrt{2}}{\max |\mathbf{u}| k_{\max}}, \quad (143)$$

and the realized CFL (40) is $\Delta t_k \max |\mathbf{u}_{k+1}| k_{\max}/2\sqrt{2}$.

A resolution margin. A structure that concentrates below the grid scale also raises $\max |\mathbf{u}|$, and so looks like a CFL problem; but a smaller step only delays it. To tell the two apart the run also records the fraction of the dealiased energy $\frac{1}{2} \sum |\hat{u}|^2$ carried by $|\mathbf{k}| > \frac{2}{3} k_{\max}$. It stays small while the field is resolved and rises as energy piles up against the 2/3 cutoff. A realized CFL excursion with a small tail fraction is a time-step problem; one preceded by a rising tail fraction is a resolution problem.

A.3 Solenoidality

The step contributes nothing of its own to the continuity residual. Every contribution to R_i — convective, subgrid, and sponge alike — has had P_{il} applied, and $k_i P_{il} = k_l - k_l k^2/k^2 = 0$, so contracting (137) with k_i gives $k_i R_i(\hat{u}) = 0$. Each stage of (138) is then divergence free, and the divergence of each mode is carried through the step unchanged,

$$k_i \hat{u}_i^{(n+1)} = k_i \hat{u}_i^{(n)}. \quad (144)$$

Whatever divergence enters the field is therefore neither amplified nor damped by the step, and if (102) holds on input it holds on output. The residual the run log records is therefore entirely the boundary clamp's, when the clamp is enabled (§5.10), and not the time stepper's.

A.4 Dealiasing

The 2/3 mask of §5.4 is applied to the incoming field before the first stage. No stage then leaves the retained band: the convective, subgrid stress, and sponge terms are masked after their forward transforms, the projection P_{il} is diagonal in $\mathbf{k}$, and each stage argument in (138) is a linear combination of band-limited fields. The mask is applied once more to $\hat{u}^{(n+1)}$, which changes nothing in exact arithmetic but makes explicit that no content reaches the band the next scaling step will stretch.